

Steel Manufacturing Operations Analytics

Management briefing | synthetic six-month simulation

Pressing - Spot Welding - Machining - Inspection - Packaging

Production attainment	Scrap rate	Pooled OEE
90.8%	2.5%	74.4%

Key findings

Production attainment is 90.8%; aggregate quantities count process-stage completions, not shipped products.

Machining is the simulated capacity-constraint candidate at 80.6 good units per scheduled hour, with parallel machines pooled.

WELD_01 has the lowest OEE (66.8%). Check time losses, speed and scrap before deciding why.

Setup accounts for 40.7% of attributed lost time; the top three categories account for 72.9%.

The two largest defect categories account for 47.7% of recorded scrap. Shift differences cannot prove a team effect because product mix changes and each team stays on one shift.

Recommended investigations

Check the lowest-output operation: review cycle times, product mix and all parallel machines before suggesting more equipment.

Review the largest downtime categories with maintenance; collect event-level start/end times and distinguish waiting from failures.

Standardize defect reporting and separate inspection discoveries from the operation that caused the defect.

Monitor weekly machine-level OEE and attainment with agreed metric definitions; validate proposed interventions using a baseline.

Interpretation limits

All observations are synthetic. Machine-shift records are independent; there is no part genealogy, WIP or flow-conservation model. Stage counts must not be interpreted as shipped output. No causal or financial impact is claimed.